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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (Biotechnology) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21BT701 - IMMUNOLOGY AND IMMUNOTECHNOLOGY

20BT701/17BT701 - IMMUNOLOGY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) Explain the humoral immune response, focusing on the structure and function of antibodies.	10	L*1	1	1
	b) Describe the different classes of immunoglobulins and their roles.	10	L1	1	1
2.	a) Explain the structure and role of MHC class I molecules.	10	L2	1	1
	b) Classify antigens based on the following characteristics: i) Antigenic determinant. ii) Cell recognition type, iii) Origin of antigens.	10	L2	2	1
3.	a) Define artificial immunity.	10	L1	2	1
	b) Explain the various types of vaccines and their application. i) Recombinant vaccines ii) DNA vaccines iii) Subunit vaccines	10	L1	2	1
		Unit – II			
4.	a) Describe the immunological factors behind the rejection of grafts.	10	L1	3	1,4
	b) Explain the various graft types and what influences graft survival.	5	L1	3	1,4
	c) Explain how immunosuppressive medications function to stop transplant rejection.	5	L1	3	1
5.	Distinguish between autoimmune disorders and hypersensitivity reactions. Give examples and discuss how these two conditions could overlap.	20	L2	4	1,2
6.	a) Discuss the applications of stem cell transplantation. Add a note on its ethical issues.	10	L1	4	1,2
	b) Discuss the mechanisms by which HIV infection leads to impaired immune function.	10	L1	4	1,2
		Unit – III			
7.	Elaborate on the different types of immunodiffusion techniques. Discuss their applications in clinical diagnosis.	20	L2	5	1,2,4
8.	Discuss the principle of radioimmunoassay (RIA). Explain its advantages and limitations compared to other immunoassays.	20	L1	5	1,3,4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
